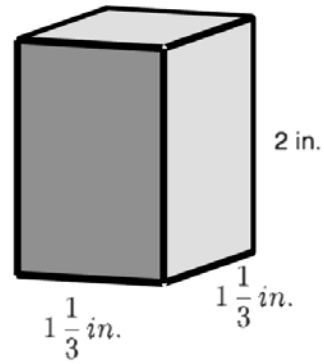
Grade 6 Accelerated | Name______Answer Key____
Unit 8 | Date____________________
Lesson 2 Homework | Period___________________

A rectangular prism is shown with units measured in inches.



In questions 1–4, find the volume using cubes with a side length of $\frac{2}{3}$ in.

1. Determine how many cubes will fit along the length, width, and height.

$1\frac{1}{3} \div \frac{2}{3} =$ ___2___ cubes will fit along the length.

$\frac{4}{3} \div \frac{2}{3} = {}^{2}\frac{4}{\cancel{3}} \times \frac{\cancel{3}}{4}{}_{1} = 2$

$1\frac{1}{3} \div \frac{2}{3} =$ ___2___ cubes will fit along the width.

$2 \div \frac{2}{3} =$ ___3___ cubes will fit along the height.

$2 \div \frac{2}{3} = \cancel{2} \times \frac{3}{\cancel{2}} = 3$

2. A total of ___12___ cubes will fit within the rectangular prism.

3. The volume of each cube is $\frac{8}{27}$ in^3. Multiply the number of cubes needed to fill the prism by the volume of each cube in order to find the volume of the prism. Include units.

$4\,\cancel{12} \cdot \frac{8}{\cancel{27}\,9} = \frac{32}{9} = 3\frac{5}{9}$ in^3

4. Instead of using cubes, Kaia uses the formula $V = l \times w \times h$. Fill in the spaces below to complete her work. Include units.

$V =$ ___$1\frac{1}{3}$___ $\times$ ___$1\frac{1}{3}$___ $\times$ ___2___

$\frac{4}{3} \times \frac{4}{3} \times \frac{2}{1} = \frac{32}{9}$

$V =$ ___$3\frac{5}{9}$___ in.^{3}

A baker is packaging her cupcakes into single serve boxes for a special event. Each cube-shaped box has side lengths 3.5 inches (in.). The smaller boxes are then packed into a larger box with dimensions of 10.5 in. $\times 7$ in. $\times 7$ in.

5. How many single serve cupcakes can be packed into the larger box?

$10.5 \div 3.5 = 3$

$7 \div 3.5 = 2$ 12 boxes can be packed into the larger box.

$7 \div 3.5 = 2$

6. Use the number of cupcakes that can be packed in the larger box to determine the volume of the box. Include units.

$(3.5 \times 3.5 \times 3.5) \times 12 = 42.875 \times 12 = 514.5$ in.^{3}

Owen and Noah are calculating the volume of a rectangular prism with the dimensions of 2 cm $\times$ 2 cm $\times$ 1 cm using different size cubes. Noah uses cubes with side lengths lengths $\frac{1}{3}$ cm and Owen uses cubes with side lengths $\frac{1}{6}$ cm.

7. Who will need more cubes to fill the larger prism? Justify.

$2 \div \frac{1}{3} = 6$		$2 \div \frac{1}{6} = 12$	
$2 \div \frac{1}{3} = 6$	108	$2 \div \frac{1}{6} = 12$	864
$1 \div \frac{1}{3} = 3$	cubes	$1 \div \frac{1}{6} = 6$	cubes

Owen will need more cubes because his cubes are smaller.

8. What is the volume of the rectangular prism? Include units.

$2 \times 2 \times 1 = 4$ cm^3

9. If they had used a cube with side lengths $\frac{1}{4}$ cm instead, would they have found the same volume of the rectangular prism?

Yes because the dimensions of the prism didn't change.

10. An aquarium in the shape of a rectangular prism is being filled with water. The dimensions of the aquarium are 48.5 inches (in.) by 18.25 in. If the height of the water is 15.75 in., what is the volume of the water in cubic inches?

$48.5 \times 18.25 \times 15.75 = 13{,}940.71875$ or $13{,}940\frac{23}{32}$